

of the centre frequency, which is in the range of tens to hundreds of megahertz, an ultra-wide-band HPM radiates microwave energy over a bandwidth in the range of hundreds of MHz to several gigahertz.

Ideally, HPM weapon systems could be used to replace precision-guided munitions to disable or destroy high-value targets or installations located in populated areas with minimised risk of human casualties.

Other application under consideration is suppression of enemy air

are effective only on a given class of targets that would absorb the frequency emitted by the system. Therefore in the case of a narrow-band HPM, knowledge of absorption by the target material as a function of frequency and aspect angle is an advantage. Frequency absorption data may be generated by scanning the target with a tunable low-energy microwave source and evaluating the reflected signals for missing frequencies. Missing frequencies in this case are those that are heavily absorbed by the target material.

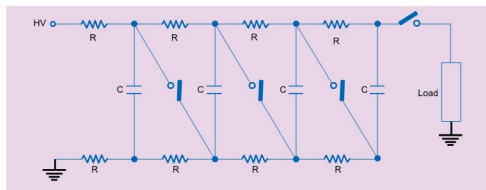


Fig. 2: Marx bank power supply arrangement

defence. Most air defence systems are highly susceptible to HPMs. Surface-to-air missiles contain highly-sensitive guidance systems and fire control is usually radar-controlled. Environmentally-dangerous targets such as chemical and biological weapon-production facilities could be targeted more effectively by HPMs without the fear of releasing deadly toxins into the atmosphere.

Types of HPM weapons

As mentioned earlier, HPM weapons are generally categorised as narrow-band HPM and ultra-wide-band HPM. Narrow-band HPM weapons produce an HPM output in a narrow band of frequencies with the bandwidth equal to only a few per cent of the centre frequency. These weapons are capable of generating relatively-higher output power levels as compared to ultra-wide-band HPM weapons.

Narrow-band HPM systems

Narrow-band HPM weapons have the advantage of better transmission characteristics and fewer problems with fratricide. Limitations include their susceptibility to countermeasures such as target hardening and prior knowledge of the threat required to choose the optimum microwave frequency.

Ultra-wide-band HPM weapons, on the other hand, radiate over a broad frequency range but deliver comparatively less microwave energy at specific frequencies. While, narrow-band HPM weapons are capable of defeating only well-defined targets or class of targets, ultra-wide-band weapons are intended for use against a wide range of targets.

Ultra-wide-band HPM weapons do not require any prior information on the target's absorption characteristics and provide a broad capability range. Due to comparatively lower radiated microwave power and poorer transmission characteristics than

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